Design & Analysis of Complaint Prosthetic Digit

A CAPSTONE PROJECT REPORT

Submitted by

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Aniruddh Dalvi & Aditya Singh

DECLARATION

"I/We hereby declare that this project work is my/our own work and that, to the best of my/our knowledge and belief, it contains no other material or previously published material written by another person which has been accepted for the award of any other degree of the university or other institute of higher learning, except where due acknowledgment has been made in the project report".

Date	Name	Roll No	Signature
	Aniruddh Dalvi	F008	
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ABSTRACT

Prosthetics have played a trans-formative role in the lives of individuals with limb loss or limb impairment. Over the years, prosthetic devices have evolved from rudimentary wooden limbs to cutting-edge, bio-mechanically advanced marvels. They offer improved mobility, dexterity, and comfort, enabling users to engage in various activities, including sports and recreation. Prosthetics have not only restored physical capabilities but also contributed to social inclusion and psychological well-being, empowering individuals to lead fulfilling lives post-amputation. This project analysed the design of a prosthetic finger based on compliant mechanism. This mechanism was chosen as it offers numerous advantages, including reduced part count, simpler manufacturing, and improved reliability due to fewer joints and moving parts. The to design a prosthetics finger various factors need to be taken into account. This project tried to find a much more versatile design for a digit which can be incorporated in everyday life.

The kinematics of the digit were studied and the various loads which can act upon it. It was found that the modelling and simulation of a compliant mechanism based system is tough and requires a more in-depth study.

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Abbreviations

ABS Acrylonitrile Butadiene Styrene v

dp Distal Phalanx 4

dm Middle Phalanx 4

pp Proximal Phalanx 4

DIP Joint Distal Interphalangeal Joint 4

PIP Joint Proximal Interphalangeal Joint 4

MCP Joint Metacarpophalangeal Joint 4

DOF Degree of Freedom 4

FDM Fused Deposition Modeling 3

CAD Computer Aided Design 7

IGES Initial Graphics Exchange Specification 7

1 Introduction

1.1 Overview

Prosthetics are artificial devices used to emulate a missing or a non functioning body part. They are pivotal in improving the lives of people who have experienced limb loss due to an accident or congenital anomalies. The primary aim of these appendages is to restore the function and aesthetics of the missing body part such that the individual is able to regain semblance of their of former abilities or even enhance them. Prosthetics are categorised into various fields; Limb Based Prosthetics, Dental Prosthetics, Ocular Prosthetics or Cranio-facial Prosthetics. Due the recent advancement in technology the manufacturing of custom-made prosthetics are more viable for various individuals.

1.2 Problem Statement

Limb based prosthetics are the most common known application of prosthesis. Human limbs, mainly digits play a key role in day-to-day life. The loss of their functionality can hinder and reduce the dexterity of the individual. The artificial digits are customized to function as a normal human digit.

1.3 Need of Study

The issue with the existing solutions is that the products which provide high level of mobility have a high premium and are proprietary. So, it is inaccessible to the general populous. There exist solutions which are open source and practical for daily use. These are usually Cable Controlled Tendon Driven fingers. These systems increase the dexterity but are highly delicate and can only be used for low load bearing applications.

1.4 Objective

The aim of this research was to design a prosthetic digit based on flexure joints for real world applications which can be produced as a single part object through additive manufacturing. This digit should be able to provide a basic level of dexterity and replicate the functionality of a actual finger. The use of flexure joints reduces the number parts and their potential points of failure.

2 Literature Review

This project required research into a variety of topics like prosthetics, compliant mechanisms, bio-mechanics and additive manufacturing. It was required to amalgamate all the topics for this study.

2.1 Prosthetics

The existing solutions had to be analysed and their mechanisms were studied. The existing mechanisms used to actuate prosthesis are categorised into various categories based on their working and application. *Cosmetic Covers* are shells made to match the appearance of human fingers, these have no other application other than aesthetics. *Myo-Electric Digits* use electromyographic signals from the muscles of the residual limbs via electrodes. These signals are translated for the actuation of the finger. The systems are good at mimicking human fingers but they are complex which drives their cost up. *Cable Driven Digits* use cables and pulleys to transmit motion from the wrist to the finger. *Tendon-Actuated Digits* have a rigid member attached to the wrist which acts as a tendon to flex or extent the finger. Like the Cable Driven systems these can fail if it experiences impact or a heavy load.

2.2 Compliant Mechanisms

A compliant mechanism is a type of mechanical structure or device that achieves its functionality through the flexibility and deformation of its components, rather than relying on traditional rigid components and joints. Compliant mechanisms are designed to take advantage of the inherent elasticity and flexibility of materials to perform specific tasks or functions. The conversion of conventional joints to lateral joints is not a streamlined process [1].

2.3 Anisotropic Materials

These type of materials have differing properties in to various direction. Advancements in additive manufacturing have opened up new possibilities in creating anisotropic materials with intricate structures and tailored properties[2]. 3D printing techniques can be used to design and fabricate materials with precisely controlled anisotropic characteristics, enabling customized components for various applications.

3 Methodology

3.1 Material

ABS was chosen as the material as it has good impact absorbing properties.

3.1.1 Anisotropic Properties

The prototype for this project is 3D printed using Fused Deposition Modeling (FDM), the properties of the material are varying in direction of the extrusion. In this case the material properties vary in the transverse axis and the properties are uniform in the orthogonal plane to the axis. This occurs as the adhesion between the different layers of the material is imperfect, creating a potential point of failure on that axis. The

The anisotropic properties of ABS where taken from Table 5[2].

Table 3.1: Anisotropic Material Properties of ABS

Property	Direction	Value
ρ (kg/mm^3)	x, y, z	1.03e-06
E (MPa)	E_x	2432.29
	E_y	2432.29
	E_z	2367.10
ν	ν_{xy}	0.34
	ν_{yz}	0.24
	ν_{zx}	0.24
G (MPa)	G_{xy}	980.76
	G_{yz}	830.47
	G_{zx}	830.47

The Hooke's Law can be applied here to find the relation in stress and strain of the material.

$$\{\sigma\} = [c]\{\epsilon\} \tag{3.1}$$

$$\{\epsilon\} = [s]\{\sigma\} \tag{3.2}$$

here,

$$[s] = [c]^{-1} \tag{3.3}$$

The equation Eqn 3.2 can be written as,

$$\begin{bmatrix} \epsilon_{xx} \\ \epsilon_{yy} \\ \epsilon_{zz} \\ \gamma_{xy} \\ \gamma_{yz} \\ \gamma_{zx} \end{bmatrix} = \begin{bmatrix} \frac{1}{E_x} & -\frac{\nu_{xy}}{E_y} & -\frac{\nu_{zx}}{E_z} & 0 & 0 & 0 \\ -\frac{\nu_{xy}}{E_x} & \frac{1}{E_y} & -\frac{\nu_{zy}}{E_z} & 0 & 0 & 0 \\ -\frac{\nu_{xz}}{E_x} & -\frac{\nu_{yz}}{E_y} & \frac{\nu_1}{E_z} & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{1}{2G_{yz}} & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{1}{2G_{xz}} & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{1}{2G_{xy}} \end{bmatrix} \times \begin{bmatrix} \sigma_{xx} \\ \sigma_{yy} \\ \sigma_{zz} \\ \tau_{xy} \\ \tau_{yz} \\ \tau_{zx} \end{bmatrix}$$

This matrix is used to analyse the sress-strain analysis on the

3.2 Model

3.2.1 Anatomy of Digit

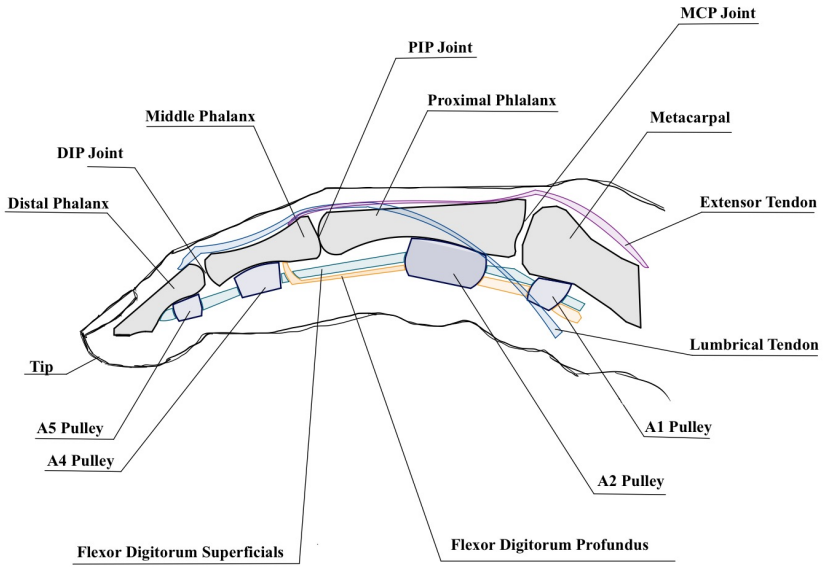


Figure 3.1: Human Finger Anatomy

The finger is made up of three links : Distal Phalanx (dp), Middle Phalanx (dm) and Proximal Phalanx (pp); The phalanges or links are connected via Distal Interphalangeal Joint (DIP Joint) and Proximal Interphalangeal Joint (PIP Joint). The whole finger is attached to the palm or the metacarpal Metacarpophalangeal Joint (MCP Joint). These joints are constrained by ligaments such that each joint acts as a revolute joint and has a single rotational Degree of Freedom (DOF) and the Volar Plate ligaments prevent the finger from bending backwards. The finger assembly is actuated by flexor tendons and extensor tendons which help the bend inwards and extend outwards respectively. The flexor tendons are routed via pulleys which help with the finer motor motions.

3.3 Numerical Model

3.3.1 Assumptions

It can be assumed that the finger is a bar of uniform cross-section with three nodal hinges [3]. The human finger is anisotropic in nature by it was taken to be isotropic and brittle materials. Looking at the figure (Fig:3.1) it can be assessed that the potential point of failure of the the fingers is at the joints as strength of ligaments rather than the cortical bone which has higher flexure strength.

3.3.2 Theoretical Calculation

The linkage diagram of the finger can be represented as the following. The phalanges are drawn as four-bar chain mechanisms and the pulleys and tendons (Fig: 3.1) are replaced by springs as their functionality is similar.

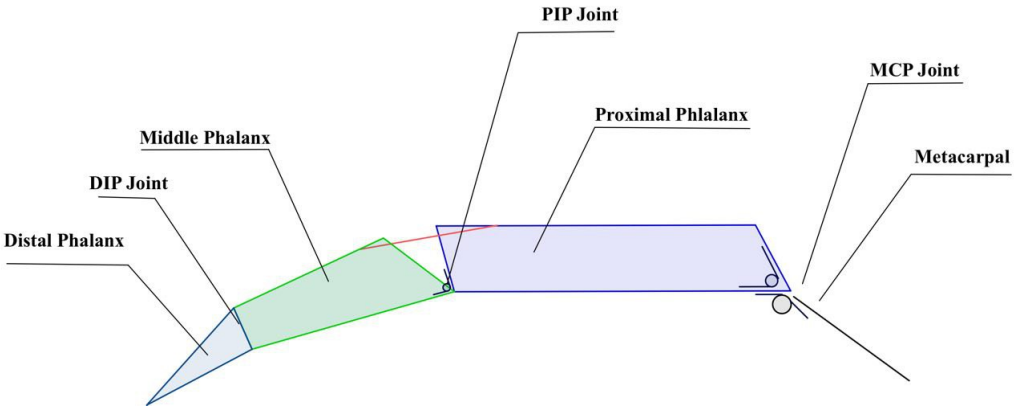


Figure 3.2: Linkage Diagram of Finger

The diagram can be simplified into a kinematic beam diagram. This is done to

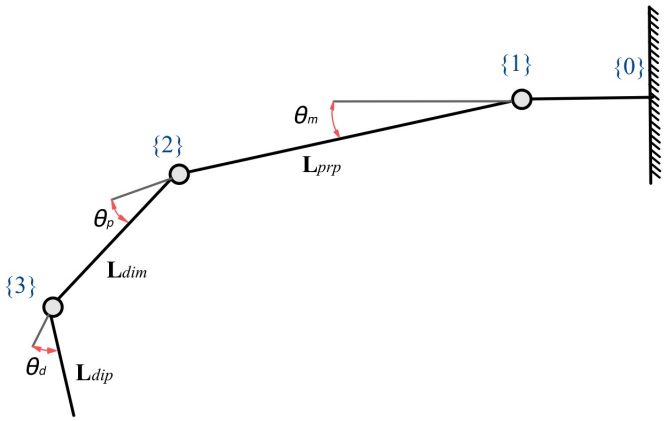


Figure 3.3: Kinematic Diagram of Finger

The 2-D kinematic diagram (Fig:3.3) of the finger can be used to visualize the motion of the finger for extension and flexture. The ground frame $\{0\}$ represents the Metacarpal. The frames $\{1\}, \{2\}, \{3\}$ are used to denote the MCP Joint, PIP Joint and DIP Joint; the joint angles are denoted by $\theta_m, \theta_p, \theta_d$ respectively. The length of the links *Distal-Phalanx*, *Middle-Phalanx* and *Proximal-Phalanx* are $L_{dip}, L_{dim}, L_{prp}$ respectively. The Denavit-Hartenberg Parameters[4] were used to obtain the common transformation matrix.

$$T_4^0 = Rot(\hat{z}, \theta_m)Trans(\hat{x}, L_{prp})Rot(\hat{z}, \theta_p)Trans(\hat{x}, L_{dim})Rot(\hat{z}, \theta_d)Trans(\hat{x}, L_{dip}) \tag{3.4}$$

This can be written as a simplified matrix as

$$[T_4^0] = \begin{bmatrix} C_{(\theta_m+\theta_p+\theta_d)} & -S_{(\theta_m+\theta_p+\theta_d)} & L_{prp} + L_{dip} \cdot C_{(\theta_m+\theta_p)} + L_{dim} \cdot C_{\theta_m} \\ S_{(\theta_m+\theta_p+\theta_d)} & C_{(\theta_m+\theta_p+\theta_d)} & L_{dip} \cdot S_{(\theta_m+\theta_p)} + L_{dim} \cdot S_{\theta_m} \\ 0 & 0 & 1 \end{bmatrix} \tag{3.5}$$

Here, $C_\theta, S_\theta \rightarrow cos(\theta), sin(\theta)$

3.3.3 Measurements

The measurements were done using a Vernier Caliper.

L_{prp}	$55mm$
L_{dim}	$30mm$
L_{dip}	$25mm$
θ_m	0° - 90°
θ_p	0° - 105°
θ_d	0° - 90°

Table 3.2: Measured Values

3.3.4 Loading Conditions

The force acts on the links depending on the position of the finger. Three cases were considered based on various grips. In this scenario the all the joint angles are zero and the load is acting axially. The load was taken to be 17% of the body mass which is 10 kg.

***Case1*: Fully Extended**

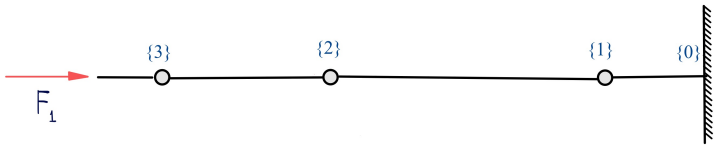


Figure 3.4: Fully Extented Finger

***Case2*: Partially Extended**

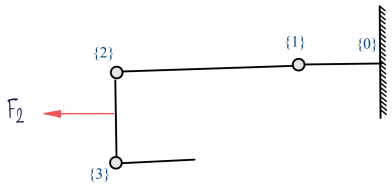


Figure 3.5: Partially Flexed Finger

***Case3*: Fully Flexed**

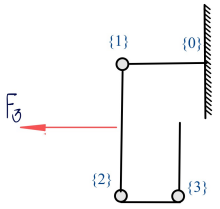


Figure 3.6: Fully Flexed Finger

3.4 Simulation

The simulations were conducted using the various analysis systems of *Ansys v.2023R2* and the Computer Aided Design (CAD) model was created using *Solidworks v.2022*. The cad model was exported in Initial Graphics Exchange Specification (IGES) format for minimal data loss.

3.4.1 Geometry

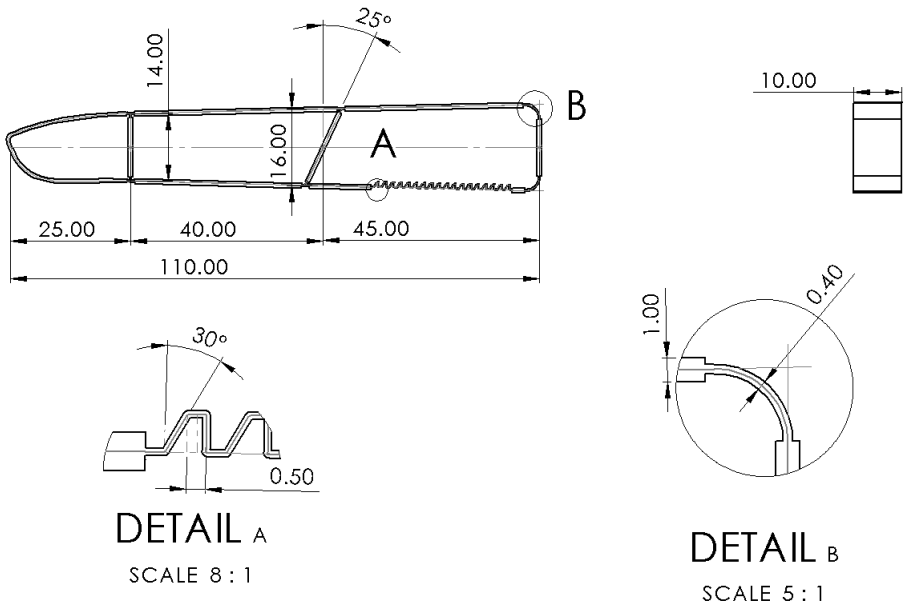


Figure 3.7: Dimensions of the Complaint Finger

The conversion of the linkage mechanism (Fig:3.2 into a compliant based linkage chain was done via trial and error. The method to convert conventional revolute joints into flexure joints was used [1].

3.4.2 Meshing

The meshing was done using program controlled method and the element size was set to 1mm.

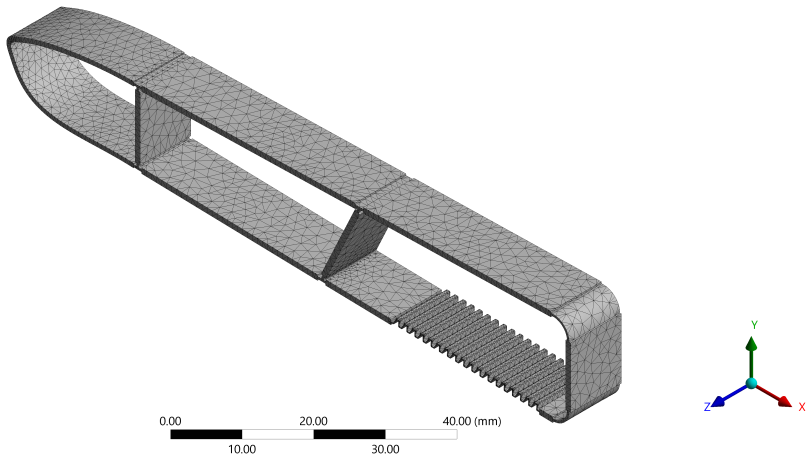


Figure 3.8: Mesh of the Complaint Finger

3.4.3 Boundary Conditions

Two cases have been considered for the boundaries

Axial Load

The end is fixed and the load is applied to the DIP Joint of 10N.

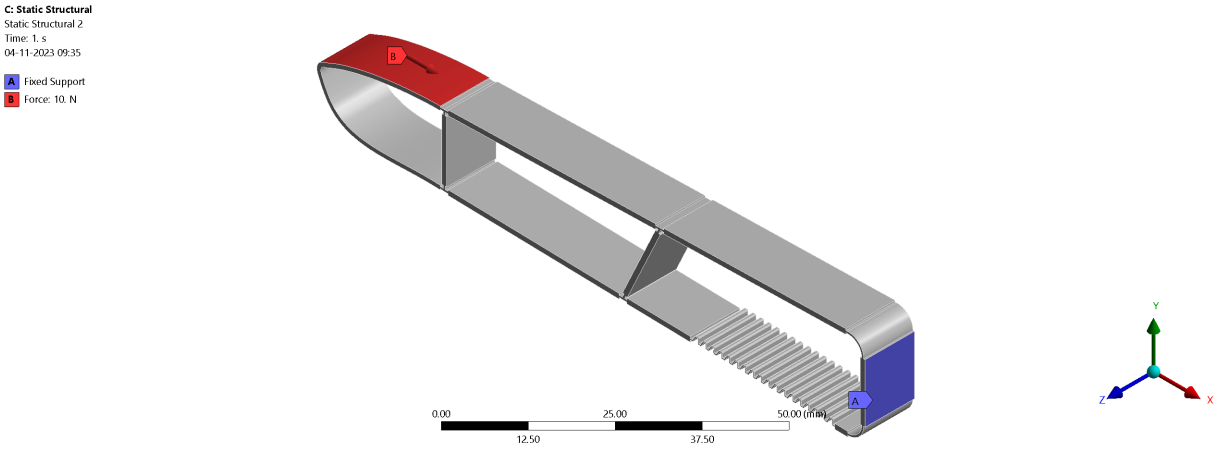


Figure 3.9: Axial Loading of Prosthetic Finger

Lateral Load

The load of 1N is applied to the lateral side.

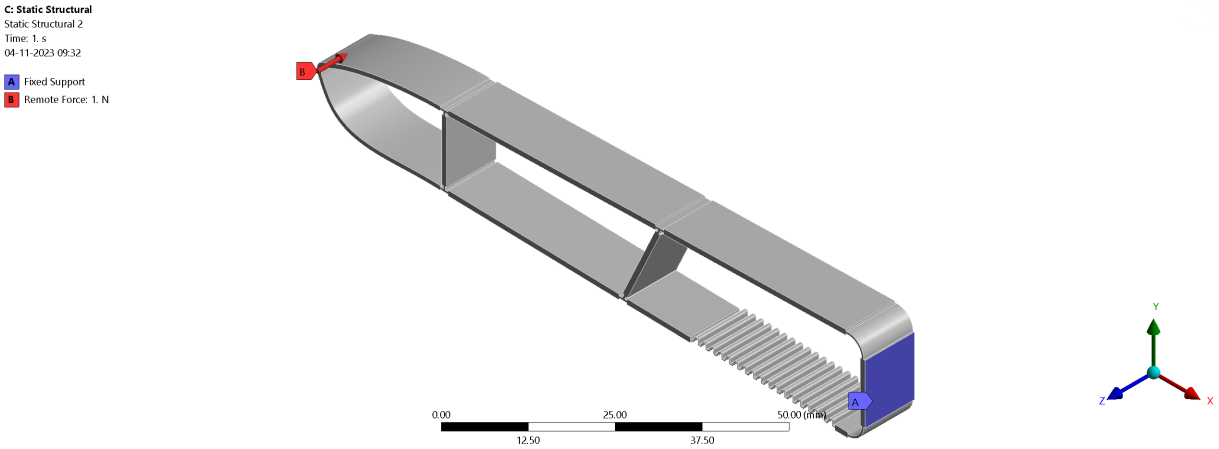


Figure 3.10: Lateral Loading of Prosthetic Finger

4 Result

4.1 Displacement

4.1.1 Lateral Loading

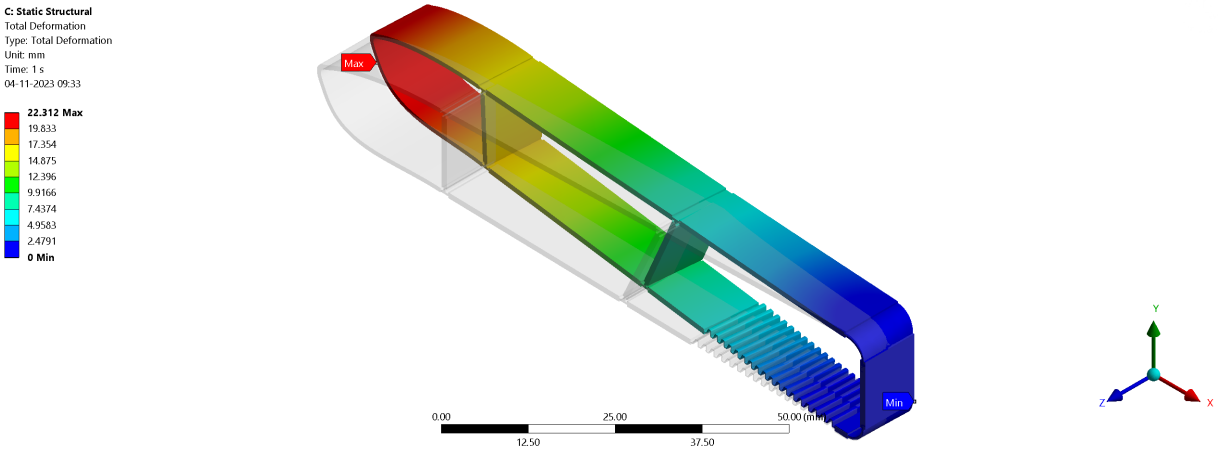


Figure 4.1: Displacement of Prosthetic Finger in lateral load

4.1.2 Axial Loading

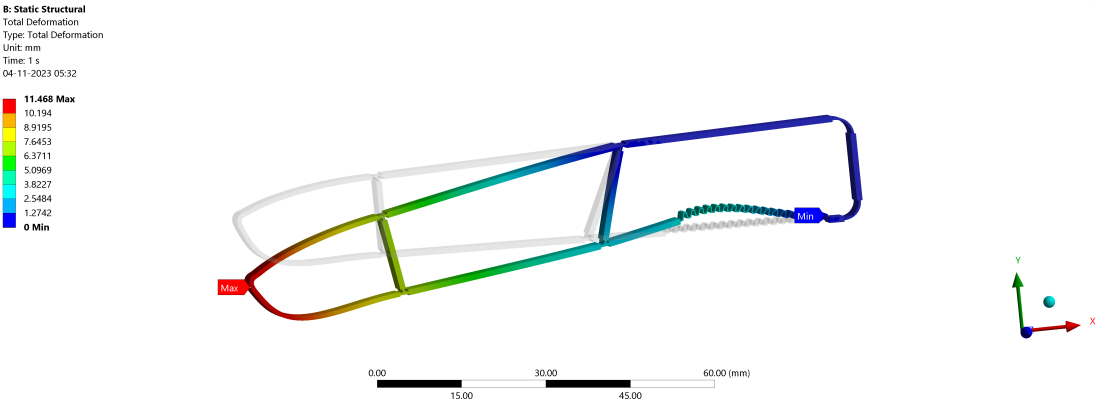


Figure 4.2: Displacement of Prosthetic Finger in axial load

Table 4.1: Shear Stress of Finger

Shear Stresses τ	Minimum(MPa)	Maximum(MPa)	Average(MPa)
τ_{xy}	-28.373	22.129	1.62e-02
τ_{yz}	-22.355	23.768	9.6e-02
τ_{zx}	-39.463	36.146	2.09e-03

5 Conclusion

The digit, in its current design, excels in handling axial loads, making it proficient in tasks that involve forces directed along its length. However, it presents limitations when subjected to lateral loads due to its relatively weaker shear modulus in the Z direction. This mechanical property makes it less suitable for applications requiring resistance to forces acting perpendicular to its longitudinal axis.

One of the notable advantages of this finger design is its ease of manufacture and prototyping. The simplicity of the manufacturing process allows for cost-effective and efficient production, which is particularly beneficial for mass production or rapid prototyping in research and development scenarios. This characteristic makes it accessible for a wide range of applications, from industrial robotics to medical prosthetics.

Furthermore, the digit's customized based on individual dimensions is a key attribute that enhances its versatility. By tailoring the finger's size and shape to match a specific user's requirements, it can be seamlessly integrated into various applications, ensuring optimal performance and user comfort. This feature is particularly valuable in fields such as orthopedics, where personalized medical devices are in high demand.

Another noteworthy advantage is the ease with which the digit can be replaced and recycled in case of failure or damage. The modular nature of the design allows for quick and cost-effective replacements, reducing downtime and maintenance costs. Additionally, the materials used in its construction can be selected with recycling in mind, contributing to sustainability efforts by minimizing environmental impact and reducing waste.

In summary, while the digit excels in axial load-bearing applications and offers ease of manufacturing and customization, it is essential to consider its limitations, especially when dealing with lateral loads. However, its replace ability and recyclability make it a practical and sustainable choice for a wide range of applications, with the potential to serve as a reliable component in various mechanical systems.

6 Future Scope

This project was primarily focused on assessing the feasibility of the prototype. By successfully demonstrating its viability, we have paved the way for its potential integration into a mass production system, such as injection molding. Looking ahead, there are several avenues for improvement and optimization that we can explore. First and foremost, in the future, this prototype can undergo further refinement and enhancement to make it more suitable for large-scale production. This could involve fine-tuning the design and engineering aspects to ensure that it can meet the requirements and standards of a high-volume manufacturing process. Additionally, the optimization of this project can extend to the development of a more streamlined and efficient process for both the design and prototyping stages. This would not only reduce the time and resources required but also enhance the overall quality of the final product. A more efficient process can involve the use of advanced CAD (Computer-Aided Design) software, automation, and cutting-edge manufacturing techniques to create a more cost-effective and precise production pipeline. In conclusion, this project has opened up exciting possibilities for the future. With further development and optimization, this prototype can potentially revolutionize the way products are manufactured, making them more accessible and efficient for a wide range of applications.

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